

NETWORK HUNTER

Complete SIH / Hackathon Prototype Documentation

Project Theme: Evidence-First Relationship Intelligence Platform

Core Definition

Network Hunter is a source-traceable relationship intelligence platform that converts fragmented investigative records into interactive relationship graphs, chronological timelines, cross-case connections, and human-verifiable insights.

Important Scope Note

This document describes a realistic hackathon prototype. It does not claim live access to police databases, CDR systems, banking systems, or classified government data. The prototype should use synthetic or authorized sample records.

1. Problem Understanding

Investigative information is often distributed across FIRs, reports, communication records, financial transactions, vehicles, locations, organizations, social/public information, and historical case records. Manual cross-referencing is slow and makes hidden relationships difficult to identify.

The actual problem is not simply drawing a graph. It is connecting fragmented records while preserving source, date, context, uncertainty, and human verification.

- Fragmented records across multiple systems
- Manual cross-referencing and slow analysis
- Duplicate or inconsistent entity names
- Difficult timeline reconstruction
- Hidden cross-case relationships
- Lack of source traceability
- Risk of false positives when records are automatically merged

2. Honest Feasibility Assessment

Area	Assessment
Hackathon prototype	Highly feasible
Graph and timeline UI	Highly feasible
Synthetic-data demonstration	Very feasible
Basic AI assistance	Feasible
Live police data integration	Not feasible without authorization
Production deployment	Requires security, legal, integration and institutional approval
Global novelty	Broad concept already exists; differentiation must come from workflow and execution

The realistic version is an investigative analytics workspace, not a complete real-time national intelligence system.

3. Core Differentiation

Existing tools already provide graph/link analysis. Therefore, do not claim that Network Hunter invented criminal-network visualization.

Defensible positioning:

Existing tools show connections. Network Hunter explains the connection, its source, its timeline, and its verification status in one investigator-friendly workflow.

- Evidence-linked relationships
- Graph and timeline working together
- Human-in-the-loop entity resolution
- Cross-case relationship discovery
- Neutral, source-grounded AI summaries
- Simple case-centric interface
- Clear verification and uncertainty indicators

4. Existing Solution Landscape

Existing solution	What it generally provides	Network Hunter opportunity
IBM i2 Analyst's Notebook	Advanced link analysis, timelines, intelligence workflow	Simpler evidence-first workflow for a focused prototype
Maltego	Entity pivots and OSINT relationship discovery	Authorized internal records, case-centric workflow and source traceability
Sentinel Visualizer	Relationship mapping and network analysis	Integrated timeline, evidence review and human verification
Neo4j	Graph database and graph analytics	Complete end-user investigation application
Gephi	General network visualization and metrics	Data ingestion, cases, evidence and timeline workflow
Existing government databases	Record storage and case management	Analytical layer across fragmented records

5. Prototype Modules

Login and Access: Basic investigator, analyst and admin roles.

Data Upload: Upload CSV, Excel or JSON sample records.

Data Validation: Check missing fields, invalid formats and duplicates.

Entity Resolution: Suggest possible matches without silently merging people.

Relationship Graph: Interactive graph of people, accounts, vehicles, cases, locations and organizations.

Timeline Analysis: Chronological events with date-range filtering.

Evidence Panel: Display source ID, date, case ID, record type and verification status.

Cross-Case Search: Find entities appearing in multiple cases.

Network Analytics: Centrality, bridge entities, clusters and shortest paths.

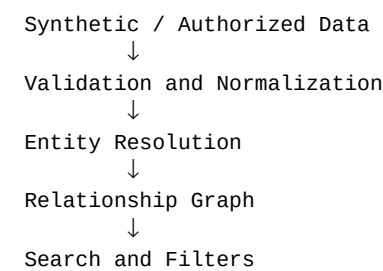
AI Assistant: Neutral summaries, connection explanations and timeline briefs.

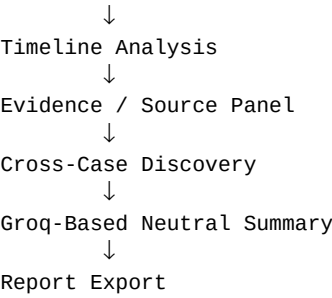
Report Export: Export graph details, timeline and summary.

6. Recommended MVP

- CSV/JSON upload
- Data validation and normalization
- Interactive relationship graph
- Search by person, phone, account, vehicle or case
- Timeline filter
- Click an edge to view source evidence
- Cross-case relationship discovery
- Verification status for uncertain matches
- Basic neutral AI summary using Groq
- Simple export/report

7. Demo Workflow





8. Suggested Synthetic Dataset

Real operational police data is restricted. Use fictional but realistic structured records.

File	Purpose
persons.csv	Person IDs, names, phone references
organizations.csv	Companies or organizations
locations.csv	Locations and coordinates/addresses
accounts.csv	Fictional bank or wallet accounts
vehicles.csv	Vehicle registration and owner references
cases.csv	Case IDs, dates and descriptions
communications.csv	Fictional communication metadata
transactions.csv	Fictional transaction records
events.csv	Time-based events
relationships.csv	Source-target relationship edges
sources.csv	Source record metadata

Label the dataset clearly: “Synthetic Data — For Demonstration Only.”

9. Relationship Record Structure

```
{
  "source_entity": "PERSON_001",
  "target_entity": "ACCOUNT_1001",
  "relationship_type": "linked_to",
  "source_id": "TXN_0042",
  "case_id": "CASE_101",
  "date": "2025-01-12",
  "verification_status": "confirmed",
  "confidence": "high"
}
```

10. Technology Stack

Layer	Recommended tool
Frontend	React + Vite + TypeScript
UI	Tailwind CSS + shadcn/ui
Graph visualization	Cytoscape.js
Backend	FastAPI + Python
Data processing	Pandas
Validation	Pydantic
Entity matching	RapidFuzz + deterministic rules

Layer	Recommended tool
Graph analytics	NetworkX
Database	PostgreSQL
AI API	GroqCloud API
AI SDK	Groq Python SDK
Authentication	JWT / basic RBAC
Charts	Recharts
Deployment	Vercel + Render/Railway
Version control	GitHub

11. Groq API Integration

Use GroqCloud for fast language-model assistance. The API key must remain in the backend and never be exposed in frontend code.

Backend flow:
Frontend → FastAPI → Retrieve relevant records → Groq API → Neutral response → Frontend

Install:

```
pip install groq python-dotenv
```

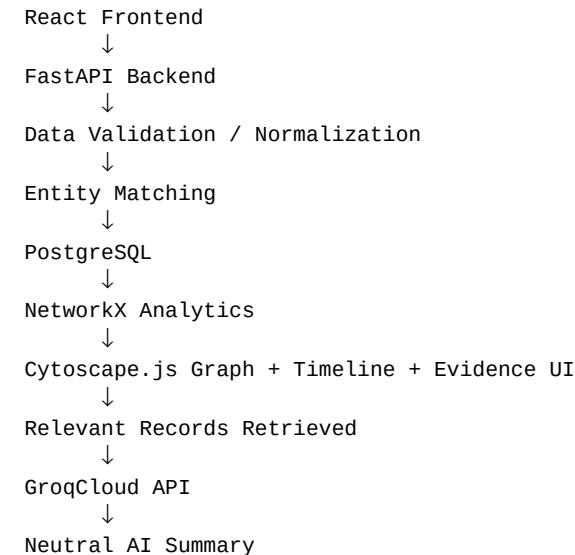
Environment:
GROQ_API_KEY=your_api_key_here

Appropriate Groq use cases:

- Case summary
- Explain a selected connection
- Extract entities from uploaded text
- Generate a chronological narrative
- Answer questions over selected records
- Draft a source-grounded report

Python should determine factual relationships; Groq should explain those facts. Do not let the LLM independently invent links or decide guilt.

12. System Architecture



13. Real-World Applications Beyond Police

- Banking fraud and anti-money laundering
- Insurance fraud detection
- Cybersecurity incident investigation
- GST and tax fraud analysis
- Corporate procurement and vendor fraud
- Telecom fraud analysis
- Customs and logistics investigation
- Healthcare claims fraud
- Supply-chain risk analysis
- Public procurement monitoring

The reusable core is relationship intelligence, not only criminal investigation.

14. SWOT Analysis

Strengths	Weaknesses
Visually strong graph and timeline demo	No real operational data
Feasible with synthetic data	Limited domain expertise
Evidence traceability and explainability	Entity matching is difficult
Reusable across domains	Security and integration complexity

Opportunities	Threats
Financial fraud, cyber, insurance and compliance expansion	Existing commercial platforms
Institutional pilot or research partnership	False positives and privacy concerns
Source-grounded AI assistance	Scope becoming too large
Modular analytical layer over existing systems	Procurement and trust barriers

15. What Not to Build in the First Version

- Live CDR integration
- Direct CCTNS/ICJS access
- Live bank transaction feeds
- Facial recognition
- Live social-media scraping
- Predictive crime or guilt scoring
- Autonomous arrest or risk decisions
- Advanced GNN models
- Nationwide production deployment
- Complex microservice architecture

16. Judge Questions and Safe Answers

Question	Suggested answer
Do you have real police data?	No. We use synthetic records because operational data requires authorization. The data model is designed for
Is this already available?	Advanced link-analysis tools exist. Our focus is a simpler, source-traceable, case-centric workflow combining g
Will AI identify criminals?	No. AI only summarizes supplied records and highlights relationships for human review. It does not determine g
Can it replace existing databases?	No. It acts as an analytical layer over existing systems and can support read-only imports or future integrations.
What is the main innovation?	Evidence-linked relationship analysis combined with timeline reconstruction and human verification.
Can it scale to other domains?	Yes. The same engine can support AML, insurance fraud, cybersecurity and corporate compliance by changing

17. Final Pitch

Network Hunter is an evidence-first relationship intelligence platform that transforms fragmented authorized records into explainable graphs, chronological timelines, cross-case connections, and human-verifiable insights. It does not replace investigators or make guilt decisions; it helps authorized analysts understand complex networks faster and with better source traceability.

Core formula:

Fragmented Data + Entity Resolution + Graph + Timeline + Evidence Traceability + Cross-Case Discovery + Human Veri